

ECM3412

UNIVERSITY OF EXETER

**COLLEGE OF ENGINEERING, MATHEMATICS
AND PHYSICAL SCIENCES**

Nature-Inspired Computation

TWO HOURS

Answer question 1, and two out of the four other questions.

Question 1 is worth 40 marks. Other questions are worth 30 marks each.

**Candidates are advised to spend FORTY-FIVE minutes on question 1
and THIRTY-FIVE minutes on each of the other questions.**

No electronic calculators of any sort are to be used during the course of this examination.

ECM3412 (2011)

Compulsory question

1. (a) Write pseudocode for *particle swarm optimisation* (PSO). For each stage, explain how the operation affects the ability of the algorithm to find a near-optimal solution.

(8 marks)

- (b) Explain how *local search* techniques differ from simple hillclimbing and why they may be better suited to solving problems with multimodal fitness landscapes.

(6 marks)

- (c) NSGA-II (Non-Dominated Sorting Genetic Algorithm-II) is known as an elitist multi-objective evolutionary algorithm. Explain how the algorithm:

- (i) differs from a single-objective evolutionary algorithm.
- (ii) proceeds from one generation to the next and is elitist in its operation.

(8 marks)

- (d) Briefly describe three desirable features of the human immune system that would also be beneficial properties of an artificial immune system.

(6 marks)

- (e) Explain the following terms as they relate to cellular automata, illustrating your answer with examples where necessary.

- i. Toroid.
- ii. State Transition Rule.
- iii. Neighbourhood.

(6 marks)

- (f) Artificial Neural Networks (ANNs) are computational techniques inspired by the human brain. Describe the key components of McCulloch and Pitts' artificial neuron and the structures they are inspired by in neurons in the human brain.

(6 marks)

(Total 40 marks)

Please Turn Over

2. (a) Evolutionary Algorithms often use a population-based approach to optimisation.

- i. Explain what this means and why this approach is often more successful than an approach based on a single individual.
- ii. Describe the main differences in algorithm execution between a population and individual-based approach to optimisation.
- iii. Briefly explain the considerations that must be taken into account when determining population size in these algorithms.

(15 marks)

(b) Each nature-inspired algorithm has a different method of varying the extent to which it *exploits* or *explores*. For each algorithm below, describe the parameters that are modified to achieve different exploitation and exploration capability and explain how varying these alters the behaviour of the algorithm.

- i. Genetic Algorithms
- ii. Particle Swarm Optimisation
- iii. Ant Colony Optimisation

(15 marks)

(Total 30 marks)

Please Turn Over

3. A bank would like to transport money between two of its branches and must pack a security van with as much money as possible. In the bank, there are 100 bags of money each of which has a known weight and monetary value (e.g., a bag predominantly filled with 2p pieces would be heavy but would have a low monetary value, in contrast to a bag predominantly filled with £50 notes). The van has a limited weight capacity (also known) and so the van must be packed with the maximum amount of money possible without overloading it.

- (a) Describe how the problem above would be tackled with a single-objective evolutionary algorithm. In your answer, discuss the representation, computation of the objective value(s), operator types and parameter settings you would use in the optimisation.

(15 marks)

- (b) Having performed the optimisation above, the bank thinks it might benefit from investing in a larger, more expensive, van. Describe how you would modify the above algorithm to provide a trade-off between the size of van required and the total amount of money that is able to be transported in one trip. In your answer discuss any changes you would make to the representation, computation of the objective value(s), operator types and parameter settings you would use.

(15 marks)

(Total 30 marks)

Please Turn Over

4. (a) Self-organising maps (also known as Kohonen networks) are neural networks that are able to learn structure from data using unsupervised learning.
- i. Explain what is meant by the word *unsupervised* in this context and how *unsupervised learning* differs from *supervised learning*.
 - ii. Show, with the aid of equations, how a self-organising map learns from an input pattern presented to it.
 - iii. Two variables involved in the learning process in a self-organising map will change over time. Explain what these variables are, how they change, and the effect this has on the learning process.

(15 marks)

- (b) Traditionally, perceptrons learn through the use of a learning algorithm (such as the delta rule), however, evolutionary algorithms (EAs) can also be used to optimise the weights within a perceptron.

- i. Describe how you would use an evolutionary algorithm to optimise the weights in a perceptron, paying particular attention to the representation, selection process, variation operators, fitness function, parameter settings and experimental design used to test your approach.

(10 marks)

- ii. The EA could also be used to optimise the weights of a *multi-layer* perceptron (MLP). Explain how you would change the EA representation to optimise the weights of an MLP and discuss the differences in capability between the perceptron and MLP networks.

(5 marks)

(Total 30 marks)

Please Turn Over

5. (a) Describe the main algorithm and, briefly, the experimental findings in the paper Di Caro G., Dorigo M. (1998) “Ant Colonies for Adaptive Routing in Packet-Switched Communications Networks” in the *Proceedings of PPSN — Lecture Notes in Computer Science* Vol. 1498, pp. 673-682.

(15 marks)

- (b) Ant colony optimisation (ACO) and particle swarm optimisation (PSO) are both known as swarm intelligence algorithms but differ in a number of aspects of their execution.
- i. Explain why intelligence might be described as an *emergent* property of swarms and give one example of such intelligent behaviour.
 - ii. Describe the differences in the types of problems that ACO and PSO are likely to be used for.
 - iii. Describe the differences in communication between individuals in the two algorithms.

(15 marks)

(Total 30 marks)

End of Paper